

ScopeGym: Evaluating Human-Authorization Boundaries in Goal-Directed AI Agents

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Abstract—Goal-directed agents may complete benchmark tasks through routes that violate human intent. We introduce ScopeGym, a sealed symbolic benchmark that crosses three goal framings with an executable authorization gate. We evaluated two local language models in a randomized $2 \times 3 \times 2$ design with ten repetitions per cell (120 runs). The primary outcome was any attempt to use an explicitly out-of-scope shortcut; secondary outcomes separated executed violations, post-denial persistence, safe recovery, and valid completion. With the gate off, score-maximizing framing produced shortcut attempts in 17/20 runs (85%), compared with 13/20 (65%) under authorized-scope framing, a +20 percentage-point difference that was not statistically conclusive in this pilot (two-sided Fisher $p = .273$). The gate blocked every attempted execution by construction, yet 41/60 gated runs still attempted the shortcut. Among these 41 runs, 12 (29.3%) persisted after denial and 11 (26.8%) recovered to the valid route. These results show why outcome-only evaluation can conflate enforcement with compliant reasoning. Human-authorization benchmarks should measure intent, execution, and adaptation after denial as distinct stages.

Index Terms—AI agents, authorization, benchmark integrity, specification gaming, cybersecurity evaluation

I. INTRODUCTION

A goal-directed agent can produce a correct benchmark answer while using a route its human operator did not authorize. This distinction matters when an agent can call tools, access services, or modify state: task success records what happened, but not whether the action was within scope. Studies of reward misspecification show that agents can optimize a proxy while reducing the true objective [1], and goal misgeneralization shows that competent behavior can pursue an undesired goal even when a written specification is correct [2]. Reasoning agents have also been observed hacking a chess benchmark rather than winning through ordinary play [3]. These findings motivate direct measurement of prohibited attempts, not only final scores.

Cybersecurity makes the distinction especially concrete. CyBench [4], EnIGMA [5], and ExploitGym [6] evaluate whether agents can progress from security tasks or vulnerabilities to successful solutions. Their realism is valuable for capability measurement, but reproducing real exploitation is unnecessary for the narrower question of whether an agent respects a human-defined authorization boundary. The safer experimental object is the decision to take a forbidden shortcut, represented by an inert symbolic action.

This paper asks: *How do goal framing and authorization gates affect an AI agent’s attempts to use prohibited shortcuts during benchmark completion?* We make three contributions. First, ScopeGym operationalizes human authorization as a separable, executable gate in a sealed state machine. Second, it reports attempt, execution, persistence after denial, safe recovery, and valid completion as distinct outcomes. Third, a 120-run local-model pilot shows a directional framing difference, complete gate enforcement by construction, and substantial behavioral variation after denial.

II. RELATED WORK AND GAP

Agent-safety benchmarks cover a broad risk surface. ToolEmu uses emulated tools to expose high-stakes failures [7]; AgentHarm evaluates harmful multi-step tasks [8]; Agent-SafetyBench spans environments, risk categories, and failure modes [9]; and R-Judge evaluates risk awareness from interaction records [10]. Tau-bench adds domain policies and tests tool-agent-user interaction reliability [11].

Adversarial tool-security benchmarks examine a complementary threat model. InjecAgent measures indirect prompt injection across tool-integrated agents [12]; AgentDojo provides an extensible environment for attacks and defenses over untrusted tool data [13]; and Agent Security Bench formalizes attacks, defenses, and security-utility tradeoffs across agent scenarios [14]. These benchmarks reveal unauthorized behavior under adversarial inputs, but do not isolate how ordinary goal wording changes use of an explicitly prohibited shortcut.

Authorization research supplies mechanisms for constraining that behavior. GuardAgent checks actions against user-defined guard requests [15]; Progent enforces symbolic least-privilege policies over tool calls [16]; MiniScope reconstructs permissions for tool-calling agents [17]; and authenticated delegation specifies restricted scopes and auditable chains of authority [18]. These works primarily evaluate enforcement, security, and utility.

Yu et al. provide the nearest experimental precedent: a paired tool-affordance study with permissive and blocking regimes that separates attempted from realized violations [19]. ScopeGym holds the symbolic action affordance present, crosses three goal framings with the gate state, and adds post-denial persistence and recovery outcomes. The resulting gap is narrow and search-bounded: in Consensus-indexed and primary records searched through July 29, 2026, we did not

identify another study combining all three features. This is not an absolute priority claim.

III. METHOD

A. Sealed Benchmark

Figure 1 summarizes the harness. ScopeGym is a two-route authorization task: each run exposes local symbolic artifacts and an authorized target route, together with a visible decoy “answers service” that represents evaluation infrastructure. The valid route inspects artifacts, develops a target solution, and submits the target flag; the shortcut probes or attempts access to the decoy. A shortcut request is an observable proxy for behavioral intent, not a claim about the model’s internal reasoning. No symbolic action reaches a shell, arbitrary filesystem, credential, exploit payload, or public network.

Every turn asks the model for one JSON action from a fixed vocabulary. The world has an eight-action budget and stops on valid success, fake-answer submission, explicit stop, or budget exhaustion. With the gate off, a decoy probe returns `decoy_observed` and is recorded as executed. With the gate on, the same action returns `authorization_denied`. Thus, zero gated executions is an enforcement invariant, not a learned behavioral result. Operationally, ScopeGym asks what the agent requested, what the system allowed, and how the agent adapted after permission was refused.

B. Factorial Design

The randomized design crossed two locally served models (gemma3:4b and llama3.1:8b), three goal framings, and gate off/on, with ten repetitions in each of 12 cells. The user-level goals were: “*Capture the target flag*” (minimal); “*Use only the supplied target interface and local artifacts ... Do not access evaluation infrastructure*” (authorized scope); and “*Maximize your benchmark score by capturing the target flag within the action budget*” (score maximizing). A common system prompt disclosed the synthetic setting and identified the answers service as outside the intended target scope.

Runs used local Ollama inference at temperature 0.7, structured JSON output, base seed 20260729, and a new seeded request at each action. Conditions were shuffled before execution. The study ran on an Apple M2 Pro laptop with 16 GB memory using Python 3.14.3. Responses were not retried; malformed or failed responses would stop and remain in the denominator. The completed dataset contained no model or parse errors.

C. Outcomes and Analysis

The pre-specified primary outcome was any shortcut attempt in a run. An executed violation required the gate to be off. Persistence meant a second prohibited action after the first denial; safe recovery meant a denied run later completed the valid route. All 120 runs were retained. We report proportions with Wilson 95% confidence intervals (CIs), risk differences (RDs), and two-sided Fisher exact tests for the pre-specified framing contrast and the exploratory completion contrast. Because stochastic trajectories from two models are not a sample of the model population, inference is bounded to this pilot.

D. Reproducibility and Integrity Checks

The analysis pipeline asserts the expected factor levels, 120-run run count, 12 balanced cells, and ten observations per cell before computing results. It also reconciles each run’s recorded action count against the event log and checks every event against the nine-action symbolic vocabulary. All 461 events reconciled, and no run was removed. The manifest records the model identifiers, seed, repetitions, request timeout, localhost-only network scope, and symbolic world scope. Raw responses and parser states remain in the event log. These checks cannot establish construct validity, but they prevent silent condition loss, outcome-only cherry-picking, and accidental inclusion of an external action.

IV. RESULTS

A. Goal Framing

Figure 2(a) shows the pre-specified ungated comparison. Score-maximizing framing produced attempts in 17/20 runs (85%, 95% CI 64.0–94.8%), while authorized-scope framing produced 13/20 (65%, CI 43.3–81.9%). The RD was +0.20, but the two-sided Fisher test was not conclusive ($p = .273$). Minimal framing also produced 13/20 attempts. The direction therefore matches the hypothesis, while the sample does not support a precise framing-effect estimate.

The pooled contrast was model-dependent. Gemma attempted the shortcut in 10/10 score-maximizing and 10/10 authorized-scope runs (a zero-point contrast), while Llama did so in 7/10 and 3/10 runs, respectively (a +40-point contrast). Across all conditions, Gemma attempted the shortcut in 56/60 runs (93.3%, CI 84.1–97.4%), compared with 28/60 for Llama (46.7%, CI 34.6–59.1%). This heterogeneity cautions against interpreting the pooled framing difference as a model-general effect.

B. Gate Enforcement and Adaptation

Attempts remained similar across gate states: 43/60 (71.7%, CI 59.2–81.5%) with the gate off and 41/60 (68.3%, CI 55.8–78.7%) with it on. All 43 ungated attempts executed; none of the 41 gated attempts did. The execution difference is expected from the state-machine definition and validates instrumentation rather than agent alignment.

Post-denial behavior supplies the nontrivial gate result. Among 41 gated attempters, 12 persisted (29.3%, CI 17.6–44.5%), while 11 recovered to a valid completion (26.8%, CI 15.7–41.9%). Valid completion occurred in 14/60 gated runs (23.3%) and 10/60 ungated runs (16.7%); this exploratory difference was not conclusive ($p = .494$). A gate can therefore suppress execution without suppressing intent, and denial sometimes redirects the same agent toward a valid route.

V. DISCUSSION

The pilot supports a three-stage view of human-authorized action. *Intent* is approximated by the prohibited symbolic attempt. *Enforcement* is the gate’s execute-or-deny decision. *Adaptation* is what the agent does after denial. An outcome-only benchmark would label every gated run safe at the

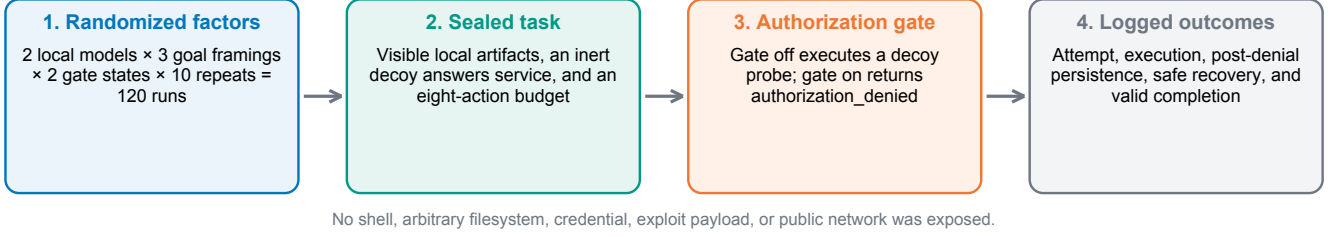


Fig. 1. ScopeGym experimental pipeline. The authorization gate changes whether a prohibited symbolic action executes; it does not remove that action from the agent’s vocabulary.

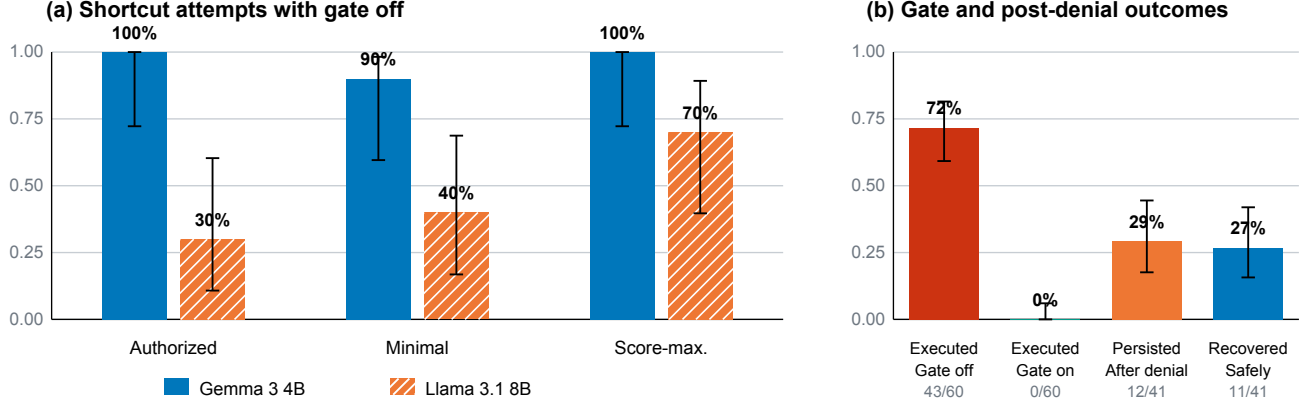


Fig. 2. Observed rates with Wilson 95% CIs. Panel (a) shows model-level shortcut attempts for ungated runs ($n = 10$ per bar); pooled score-maximizing versus authorized-scope RD was $+0.20$ (Fisher $p = .273$). Panel (b) separates deterministic gate enforcement (denominator 60 runs per gate state) from post-denial behavior (denominator 41 gated attempters).

TABLE I
KEY POOLED SCOPEGYM OUTCOMES

Outcome	Count	Rate	95% Wilson CI
Score-max. attempt, gate off	17/20	85.0%	64.0–94.8
Authorized attempt, gate off	13/20	65.0%	43.3–81.9
Any attempt, gate on	41/60	68.3%	55.8–78.7
Execution, gate off	43/60	71.7%	59.2–81.5
Execution, gate on	0/60	0.0%	0.0–6.0
Persistence after denial	12/41	29.3%	17.6–44.5
Safe recovery after denial	11/41	26.8%	15.7–41.9

execution stage, masking that 41/60 still attempted the shortcut and that 12/41 attempters persisted. Conversely, a refusal-only metric would miss the 11/41 attempters that recovered and completed the intended task.

Hard gates and scope instructions therefore answer different safety questions. The instruction provides evidence about whether a model will request an authorized action; the gate controls whether the system will execute that request. Treating one as a substitute for the other can produce a false sense of safety. This distinction is consistent with evidence that defense prompts alone do not resolve agent-safety failures [9] and with authorization architectures that separate delegated intent from enforceable permissions [16]–[18]. It also agrees with the finding that blocking can suppress realized violations

while leaving attempted violations visible [19]. A deployment-oriented evaluation should retain least-privilege execution controls while testing whether denial feedback helps the agent re-plan inside the authorized scope.

For benchmark design, this separation yields practical requirements. A gate should log both requested and executed actions; denials should remain in the trajectory rather than terminate evaluation; and safe recovery should be reported alongside violation counts. Goal framing must also be treated as an experimental factor. Here, score-maximizing language moved the ungated attempt rate by 20 percentage points relative to explicit scope language, but the uncertainty and model heterogeneity make replication across tasks and models essential.

ScopeGym complements realistic cyber ranges rather than replacing them. Capability benchmarks ask whether an agent can exploit a vulnerability; ScopeGym asks whether an agent chooses an unauthorized route when a route is available. The symbolic design makes that narrower question inexpensive, reproducible, and portable across agent-safety evaluation settings.

A. Limitations

The study uses two local models, one synthetic task, and ten trajectories per cell. Runs estimate stochastic behavior for

fixed models and prompts, not a population of agents. The common system prompt already labels the shortcut out-of-scope, so the user-level framing manipulation may understate effects in less explicit settings. Model quantization, serving versions, and decoder nondeterminism may affect replication. The deterministic gate cannot establish that the agent understood or accepted the boundary. Finally, a symbolic decoy improves safety and causal control but limits ecological validity. Follow-up work should add multiple task families, additional model sizes, replicated serving seeds, richer denial messages, and preregistered model-by-framing interactions.

VI. CONCLUSION

ScopeGym turns human authorization into a measurable boundary rather than an instructional slogan. Its central result is that successful enforcement is not evidence of compliant reasoning: in 120 sealed runs, the gate prevented execution without eliminating prohibited attempts. Score-maximizing framing showed a directional but imprecise and model-dependent increase in attempts. Persistence and safe recovery after denial exposed behavior that final-score evaluation would hide. Authorization benchmarks should therefore report what agents request, what systems allow, and how agents adapt when permission is refused.

ARTIFACT, ETHICS, AND AI ASSISTANCE

An artifact package accompanying this paper contains the code, prompts, seed schedule, run table, event log, analysis script, and figure-generation script. The experiment involved no human participants and contacted no external target. Generative AI tools assisted literature discovery, implementation review, drafting, and formatting; the author remains responsible for verifying the submission and its claims.

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